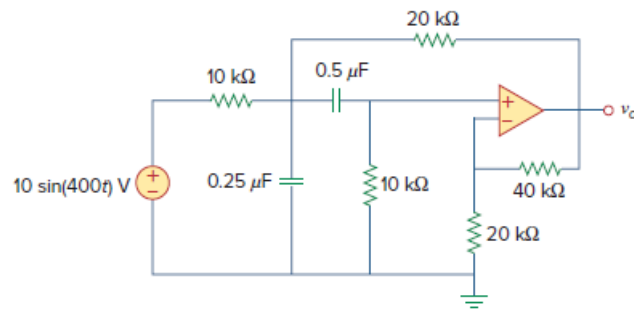


Problem

Determine $v_o(t)$ in the op amp circuit in Fig. 10.121 below.

Figure 10.121:



Step-by-step solution

Step 1 of 5

Refer to Figure 10.121 in the textbook.

Calculate the impedance of the capacitor.

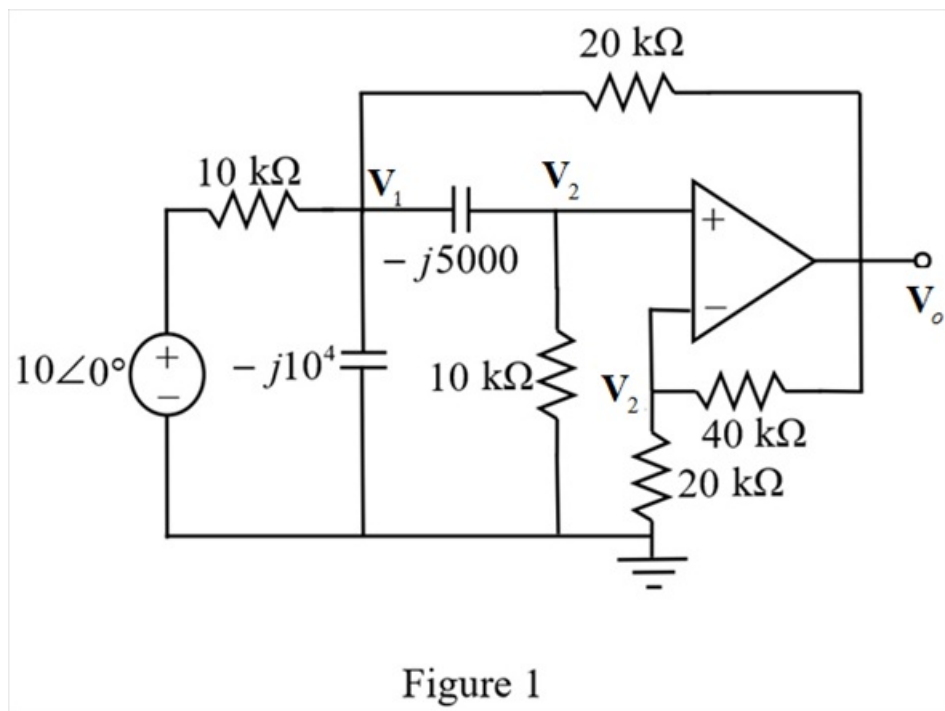
$$\begin{aligned} Z_{C_1} &= \frac{1}{j\omega C_1} \\ &= \frac{1}{j(400)(0.5 \times 10^{-6})} \\ &= -j5000 \, \Omega \end{aligned}$$

Calculate the impedance of the capacitor.

$$\begin{aligned} Z_{C_2} &= \frac{1}{j\omega C_2} \\ &= \frac{1}{j(400)(0.25 \times 10^{-6})} \\ &= -j10000 \, \Omega \\ &= -j10^4 \, \Omega \end{aligned}$$

Step 2 of 5

Draw the circuit diagram in frequency domain.



Step 3 of 5

Apply voltage division rule to calculate the voltage V_2 .

$$V_2 = \frac{20 \times 10^3}{(20 \times 10^3) + (40 \times 10^3)} V_o$$

$$= \frac{V_o}{3}$$

Apply the nodal analysis at the node V_2 .

$$\frac{V_2 - V_1}{-j5000} + \frac{V_2}{10 \times 10^3} = 0$$

$$j2(V_2 - V_1) + V_2 = 0$$

$$j2V_1 = V_2(1 + j2)$$

$$V_1 = \frac{V_2(1 + j2)}{j2}$$

Substitute $\frac{V_o}{3}$ for V_2 .

$$V_1 = \frac{\left(\frac{V_o}{3}\right)(1 + j2)}{j2}$$

$$= \frac{(1 - j0.5)V_o}{3}$$

Step 4 of 5

Apply the nodal analysis at the node $\mathbf{V}_1$.

$$\frac{\mathbf{V}_1 - \mathbf{V}_2}{-j5000} + \frac{\mathbf{V}_1}{-j10^4} + \frac{\mathbf{V}_1 - \mathbf{V}_o}{20 \times 10^3} + \frac{\mathbf{V}_1 - 10}{10 \times 10^3} = 0$$

$$\frac{j(\mathbf{V}_1 - \mathbf{V}_2)}{5000} + \frac{j\mathbf{V}_1}{10^4} + \frac{\mathbf{V}_1 - \mathbf{V}_o}{2 \times 10^4} + \frac{\mathbf{V}_1 - 10}{1 \times 10^4} = 0$$

$$\frac{j4(\mathbf{V}_1 - \mathbf{V}_2) + j2\mathbf{V}_1 + \mathbf{V}_1 - \mathbf{V}_o + 2\mathbf{V}_1 - 20}{2 \times 10^4} = 0$$

$$\mathbf{V}_1(3 + j6) - j4\mathbf{V}_2 - \mathbf{V}_o = 20$$

Substitute $\frac{(1 - j0.5)\mathbf{V}_o}{3}$ for $\mathbf{V}_1$ and $\frac{\mathbf{V}_o}{3}$ for $\mathbf{V}_2$.

$$\frac{(1 - j0.5)\mathbf{V}_o}{3}(3 + j6) - j4\frac{\mathbf{V}_o}{3} - \mathbf{V}_o = 20$$

$$\left[\frac{(1 - j0.5)(3 + j6) - j4 - 3}{3} \right] \mathbf{V}_o = 20$$

$$\left[\frac{(1 - j0.5)(3 + j6) - j4 - 3}{3} \right] \mathbf{V}_o = 20$$

$$\mathbf{V}_o(3 + j0.5) = 60$$

Step 5 of 5

Simplify $\mathbf{V}_o$ further:

$$\mathbf{V}_o = \frac{60}{3 + j0.5}$$

$$= 19.46 - j3.24$$

$$= 19.73 \angle -9.46^\circ \text{ V}$$

Write the expression for v_o in time domain.

$$v_o = 19.73 \sin(400t - 9.46^\circ) \text{ V}$$

Thus, the output voltage is $\boxed{19.73 \sin(400t - 9.46^\circ) \text{ V}}$.